

From 2016 Energy Orchestration to 2026 Agentic Systems

Why the Cost of Learning Must Decrease While Decision Quality Increases

A short white paper expanding the core principle of Energy Efficiency as a Service (EEaaS)
for today's AI agents, local-first tools, and hybrid on-device/cloud workflows

1. The Enduring Constraint

In 2016 the central observation was simple: smartphones are energy-constrained systems whose power behaviour is chaotic and multi-component. Continuous fine-grained measurement itself consumes scarce resources. The same physics still governs every modern AI agent stack.

Today's agents—whether running local SLMs, tool-using LLM pipelines, or multi-agent teams—face an identical tension. Context windows, tool calls, memory retrieval, model escalations, and continuous sensing all cost energy, latency, tokens, and money. The more an agent tries to “know everything all the time,” the more expensive it becomes. The only sustainable path is a learning loop in which the **cost of acquiring the next unit of useful information falls** while the **quality of the decisions that information enables rises**.

2. The Original Virtuous Cycle (EEaaS)

The 2016 EEaaS architecture proposed that participating devices should not stream raw telemetry forever. Instead they report selectively, the cloud learns which signals actually matter (Key Energy Indicators), and the devices progressively report less. Over iterations the monitoring overhead shrinks while the quality of power-management decisions improves. This is meta-optimization: the measurement system itself becomes more efficient as its models mature.

That principle maps directly onto today's agent workflows.

3. Mapping to Today's AI / Agent Workflows

Selective sensing and adaptive telemetry. Modern agent runtimes still default to over-logging. A 2026 equivalent of Key Energy Indicators is a set of *Key Decision Indicators* (KDIs): the minimal set of observations (tool outcomes, user corrections, confidence signals, context hashes) that actually change future policy. Agents should start broad, then prune logging and memory writes as the value of each signal class is learned.

Local-first with intelligent escalation. On-device models handle the majority of routine work. Only when local confidence is low, or when a KDI threshold is crossed, does the agent escalate to a larger cloud model or external tool. The escalation policy itself is learned and tightened over time, exactly as EEaaS reduced cellular uploads in favour of Wi-Fi / charging windows.

Memory and context as energy budgets. Long-term agent memory is not free. Writing, embedding, retrieving, and re-ranking all cost compute and tokens. The orchestration layer should treat memory operations as first-class energy (and token) consumers and learn which memories actually improve future decisions. Low-value memories are compacted or discarded; high-value patterns are retained and even shared across agents under privacy constraints.

Collective learning without continuous centralisation. Participatory sensing in 2016 became federated / privacy-preserving learning in 2026. Aggregated patterns of successful tool sequences, failure modes, and cost-quality trade-offs can improve the global policy while individual agents keep raw traces local. The orchestrator's job is to distribute improved policies and updated KDI definitions, not to hoard every observation.

4. The Modern Virtuous Cycle

An agent system optimised for decreasing learning cost looks like this:

- **Phase 1 – Broad but cheap observation.** Capture a wide but low-resolution set of signals (tool success/failure, latency buckets, user accept/reject, local model confidence).
- **Phase 2 – Importance learning.** Offline or low-priority analysis ranks signals by their contribution to decision quality (feature importance, counterfactual value, regret reduction).
- **Phase 3 – Policy tightening.** Logging, memory write, and escalation rules are rewritten to focus on the high-value signals. Communication and compute budgets shrink.
- **Phase 4 – Continuous but sparse improvement.** New edge cases still generate richer traces, but the steady-state cost of the learning loop keeps falling while average decision quality rises.

The loop is the same one proposed in 2016: measure → learn importance → reduce measurement → improve decisions → repeat. Only the substrate has changed—from smartphone system calls to agent tool traces, model confidence, and memory utility.

5. Why the Principle Is More Urgent in 2026

Three forces make the original insight more critical today than in 2016:

- **Scale of cost.** A single poorly instrumented multi-agent workflow can burn orders of magnitude more energy and money than a 2016 smartphone profiler. Token and inference costs are now first-order product constraints.
- **Local-first and edge reality.** Users increasingly expect capable agents that work offline or on-device. Continuous cloud dependence is both expensive and fragile. Selective, learned escalation is the only practical compromise.
- **Compounding agent lifetime.** Agents are no longer one-shot scripts. They accumulate memory, habits, and policies over weeks or months. Without an explicit mechanism that reduces the cost of further learning, the system's own history becomes a liability.

6. Closing Principle

The durable contribution of the 2016 EEaaS work is not any particular logging API or cloud component. It is the recognition that **energy-aware (and today, token- and latency-aware) systems must optimise the learning process itself**. When the cost of learning falls while the quality of decisions rises, the system becomes sustainable. When the opposite occurs, every additional capability eventually collapses under its own observational overhead.

That principle remains the clearest design criterion for the next generation of practical, long-lived AI agents.

Based on the 2016 Uppsala University thesis "Enabling Energy-Efficient Data Communication with Participatory Sensing and Mobile Cloud" (EEaaS). Expanded for contemporary agentic systems, August 2026.